

- 19** Adrenaline is a hormone that prepares the body for rapid responses by increasing the blood flow to muscles and increasing blood sugar concentrations. Adrenaline is secreted from the adrenal glands and acts on nearly all body tissues by binding to adrenergic receptors on the surface of the cells.

Different tissues respond to adrenaline in different ways. For example, adrenaline causes smooth muscle in the airways to relax but causes contraction of smooth muscle in arterioles.

What could explain the different responses of smooth muscle to adrenaline?

- A** Smooth muscle cells in the airways and in the arterioles have different types of adrenergic receptors, which initiate different responses after binding to adrenaline.
- B** The adrenergic receptors on smooth muscle cells in the airways bind to different ligands from the adrenergic receptors on smooth muscle cells in arterioles.
- C** The concentration of adrenaline reaching target cells is dependent on the distance from the adrenal glands and determines whether smooth muscle relaxes or contracts.
- D** The binding sites of adrenergic receptors on smooth muscle cells have different tertiary structures depending on the location of the smooth muscle cells.